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Environmental Data Exploration II

Team Charter

Link to GitHub Repository: https://github.com/JoshG07/Goozman-Woock-Chu

1. Team Purpose or Mission

Our team hopes to create a detailed dashboard informing management decisions on prescribed fires and their impact on the distribution of invasive species in the area. By the end of the project, each member aims to further develop their analytical skills using Python/R as well as gain experience transforming data into useful visuals and dashboards. Our intended audience is land managers and anyone concerned with the potential impacts of prescribed fire in their local communities. We are all interested in using our skills with different tools like GIS to develop visual aids from our project, and our group has background knowledge in fires, invasive species, and biodiversity that will support our research. The final product will be of high analytical quality, clear, and professional.

2. Communication Plan and Collaborative Coding

We have a group text chat that we'll use to communicate about the project. We will communicate regularly throughout the week to review the project goals for each week and make sure we are prepared for the in-class working sessions. We all aim to respond within the day to each message/question regarding our project. We are available to meet up and work on this project during Friday afternoons outside of the provided class time. We will have a project overview Google Doc that is routinely updated and outlines all of our completed tasks and what is needed.

3. Roles and Responsibilities

Every group member looks forward to practicing the concepts taught in the R/Python sections of the classes and wants to contribute to every step of the project. By dividing the project work, every member gets experience in data preparation, exploratory analysis, dashboard development, interpretation, documentation, and presentation. By focusing on collaborative working sessions over individual work, each of us is able to share input and see the project the whole way through. Because we are evenly dividing the work, the group cannot go to the next step until every member has done their part; this will help each of us be accountable. Before moving to the next steps of the project, we will also review what was recently done and make sure each member is happy with the work and ready to progress. As we get deeper into the project, there may be more individual work opportunities, but the group will still initially meet and discuss what needs to be done each week.

4. Operating Norms or Ground Rules

We will draft an outline with a list of tasks and deadlines for each task. We will have a weekly check-in meeting to follow up on each team member’s progress on their tasks, debug together, and assign action items for the following week. We expect that if a teammate were to complete work independently, they would notify when they started making changes as well as when

they've completed their work and pushed their changes into the repository, in a timely manner. We will draft up notes for any meeting a teammate anticipated on missing, in which they should let the team know in advance. If someone anticipates missing a deadline, they should let the team know as soon as possible so we can delegate their tasks to other members of the team. We will combine our work through pushing and pulling on github, meeting up regularly to submit the work together.

5. Values and Behaviors

Being a good teammate means being respectful of each other's ideas, being responsive and communicative throughout the week, and doing a fair amount of the workload. If there is any issue of uneven work on contributions to the project one of both of the other teammates is absolutely allowed to bring this up and have a conversation about it. Workloads will be clearly divided, and members will have their tasks outlined so there will be no confusion about what is expected of each person. All of the work and communication rules have been agreed upon by each member of the group.

6) Resources and Constraints

We have access to a variety of resources from the first half of Environmental Data Exploration, including lectures and example code. If we aren't able to figure something out from that, we can consult instructors or TAs for complicated questions and utilize documentation on the internet for simple things. We are limited primarily by time and the data we have available. The goals we set are intended to create a functional product within the timeline. This, along with the given datasets, led us to choose to focus on the impact of invasive species in California. We hope to build reusable code that could look at different species in different areas. The expectation in our group is that we will be able to follow up with each other as necessary, but we will also document everything we do along the process, so anyone can see what the assumptions, goals, and outcomes were.

AI-Use Statement: AI will only be used for editing and debugging purposes, not writing actual code. We see it as a tool to supplement our knowledge, not as something to do the work for us. When AI is used, the reason will be documented in the code.

7. Decision-Making Guidelines

Decisions will be made by meeting as a group and discussing what our options are in order to come to a consensus. After all relevant information is shared and questions have been answered, we will vote for our preferred option, if we cannot come to an agreement. If there is a significant issue affecting our ability to work together that cannot be resolved on our own, we will reach out to a TA or instructor to help us proceed.

8. Goals and Objectives

A successful project will take the form of a public Github repository and a deliverable dashboard. We hope that through our repository, we can share cleaned data sets, reproducible scripts, aesthetic visualization outputs, and detailed metadata with stakeholders to view and use. We also aim to create a dashboard that concisely communicates and visualizes our research question, methods, and results to land managers and interested community members.

To evaluate data quality, we will assess the number of missing values within the data set, ensure that our naming format is uniform across data sets, and that each variable has the correct data types. Metadata will be pivotal as a check for data quality, as we match the information in our data set to the information we list in the metadata. To evaluate dashboard functionality, we will ensure that the visualizations we include are of significance and relevance to our research question. We will also make sure to include a brief summary of our research question, methodology, and findings within our dashboard. To evaluate the clarity of our findings, we will reference relevant published literature. Essential deliverables will include the repository and dashboard. Optional features will include a brief report on our findings and ArcGIS Maps to visualize our findings.

9. Schedule and Milestones Work

We plan to divide all of the work equally, with all members contributing to code, data exploration, analysis, writing, and presentation. The deadlines set below are meant to keep us on track, with everyone responsible for making sure the team is on track. We will continually think about the scope throughout and make changes as needed. Progress will be tracked primarily by the deadlines below.

- 10/20: Research Questions and Scope
- 10/26: Complete Data Cleaning and Wrangling
- 10/30: Exploratory Data Analysis
- 11/3: Calculate statistics and produce visualizations
- 11/5: Conduct spatial analysis in GIS
- 11/10: Draft of final product and dashboard development
- 11/14: Testing and revision of final product
- 11/17: Complete final product
- 11/19: Complete final presentation